Implement basic steps of machine learning : data loading,splitting,training,and testing using a simple classifier.

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```
pip install numpy pandas scikit-learn
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

```
data = load_iris()
X= data.data #features
y= data.target #labels
```

```
#Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
 X,y, test_size=0.2, random_state=42
)
```

```
#Initialise classifier
model = LogisticRegression(max_iter=200)
#train the model
model.fit(X_train , y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=200)
```

```
#Make Predictions
y_pred = model.predict(X_test)
#Evaluate performance
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
#Detailed report
print("\nClassification Report:")
print(classification_report(y_test,y_pred))
```

```
Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

```
new_sample = [[4.1,3.9,0.5,2.2]]
```

```
prediction = model.predict(new_sample)
print("Predicted class:", prediction)
```

```
Predicted class: [0]
```